

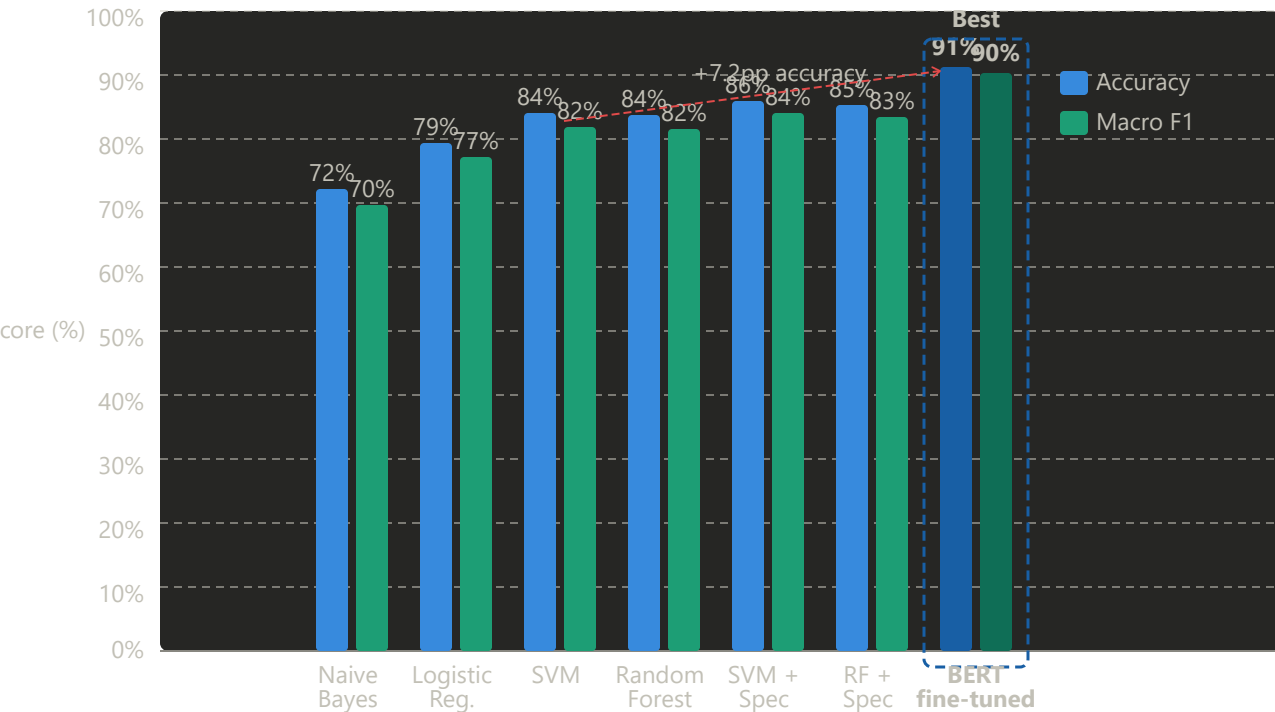


Fig. 4. Comparative classification performance (accuracy and macro F1-score) of all seven evaluated models on the five-class sentiment task using the Mobile Reviews Dataset (2025 Edition) [14]. The fine-tuned BERT model (highlighted) achieves the highest accuracy of 91.4% and macro F1 of 0.903, outperforming the best classical baseline (SVM) by 7.2 percentage points in accuracy.